

COURSE TITLE : WOOD CHEMISTRY
COURSE CODE : 3124
COURSE CATEGORY : B
PERIODS PER WEEK : 4
SEMESTER :3
PERIODS PER SEMESTER : 60
CREDITS : 4

TIME SCHEDULE

Module	Topics	Periods
I	Organic chemistry Carbohydrates Test	15
II	Physical chemistry nuclear chemistry Test	15
III	Structure Of Wood Chemistry Of Wood Constituents Wood Extractives Test	15
IV	Chemical Treatment Of Wood Bioconversion Of Wood Decomposition Of Wood Wood Analysis. Test	15
Total		60

Course Outcome :

Module	G.O	Student will be able to:
1	1	Understand the chemistry, preparation and properties of organic compounds
	2	Study the important carbohydrates
2	1	Understand physical chemistry
	2	Understand nuclear chemistry
3	1	Understand the structure of wood.
	2	Comprehend the chemistry of wood constituents
	3	Comprehend wood extraction
4	1	Understand chemical treatment of wood
	2	Understand bioconversion of wood

	3	Comprehend decomposition of wood
	4	Understand Wood analysis principles

Specific outcome

MODULE I

1.1.0 Understand the chemistry, preparation and properties of organic compounds

- 1.1.1 Write the iupac names of simple organic compounds.
- 1.1.2 Write the structural formulae of long chain and branched chain hydrocarbon.
- 1.1.3 State some properties of organic compounds.
- 1.1.4 Differentiate between saturated and unsaturated compounds.
- 1.1.5 State the preparation, phenol, ketone, aldehydes, acids and amines.
- 1.1.6 Describe the terms hydrocarbon, paraffin ole-fines and homologous series.
- 1.1.7 Explain chlorination in methane.
- 1.1.8 Describe absolute alcohol, denatured alcohol, and methylated spirit.
- 1.1.9 Describe the effects of substitutions on acidity of acids.
- 1.1.10 Describe the acidity of phenols and basicity of amines.

1.2.0 Study the important carbohydrates

- 1.2.1 State carbohydrates.
- 1.2.2 Describe the structure of glucose, starch cellulose and fructose
- 1.2.3 Describe the important reaction of glucose, starch fructose cellulose.
- 1.2.4 Describe the term terpenoids.
- 1.2.5 Describe the isolation of terpens.

MODULE -II

2.1.0 Understand physical chemistry

- 2.1.1 Explain the theory of corrosion and passivity.
- 2.1.2 Describe the causes of corrosion.
- 2.1.3 Describe the various methods for the prevention of corrosion.
- 2.1.4 Describe the term colloids, lyophilic and lyophilic of colloidal; solution.
- 2.1.5 Describe the various methods for the preparation of colloidal solution.
- 2.1.6 State the terms – gels and emulsion.
- 2.1.7 Describe the preparation of colloids (mechanical, optical, electrical)

2.2.0 Understand nuclear chemistry

- 2.2.1 Explain the effect of radioactivity and radioisotopes
- 2.2.2 Describe the radioactive series.
- 2.2.3 Describe induced radioactivity, artificial transition, and radioisotopes.
- 2.2.4 List uses of radioisotopes.

MODULE -III

3.1.0 Understand the structure of wood.

- 3.1.1 Explain structure of cell wall with figure.
- 3.1.2 State major components and minor components.
- 3.1.3 Describe the cell wall composition of soft wood and hardwood.

3.2.0 Comprehend the chemistry of wood constituents

- 3.2.1 Explain cellulose, hemicelluloses, and lignin.
- 3.2.2 Describe the composition of cellulose.
- 3.2.3 Describe the reactions of cellulose with water, alkali and acid.
- 3.2.4 Illustrate the structure of cellulose and describe micro fibril structure of cellulose.
- 3.2.5 Describe the properties of cellulose.
- 3.2.6 Distinguish between alpha cellulose, beta cellulose, gamma cellulose, holo cellulose, cross and be van cellulose.
- 3.2.7 Differentiate between cellulose and hemicelluloses.
- 3.2.8 Describe the nature of hemicelluloses.
- 3.2.9 Explain the composition of lignin.
- 3.2.10 Describe pectin.

3.3.0 Comprehend wood extraction

- 3.3.1 Explain the extraction of various components in wood.
- 3.3.2 Lists wood extractives and describe the location of extractives in the tree.
- 3.3.3 Describe the methods of extraction of extractives such as steam distillation, solvent extraction.
- 3.3.4 Describe terpenes and their derivatives tropolones, phenolic compounds such as flavanoids, stilbenes, quinones lignans, and tannins.
- 3.3.5 Describe the influence of extractives of wood.
- 3.3.6 Describe the effect of extractives on durability of wood.
- 3.3.7 Describe the effect of extractives on gluing characteristics.
- 3.3.8 Describe the effect of extractives on surface coating of wood.
- 3.3.9 State discoloration of wood.

MODULE -IV

4.1.0 Understand treatment of wood

- 4.1.1 Define Chemical treatment of wood
- 4.1.2 Describe the action of water on wood
- 4.1.3 Describe the relationship between the moisture content and strength of wood
- 4.1.4 Describe the relationship between moisture content and electrical conductivity of wood
- 4.1.5 Describe the action of aqueous solution of inorganic salt on wood

- 4.1.6 Describe the alkaline degradation of wood (soft wood and hard wood)
- 4.1.7 Describe oxidation and bleaching of wood
- 4.1.8 Describe the effect of chemicals on wood and chemical resistance of wood
- 4.1.9 Distinguish between chemical resistance of soft wood and hard wood
- 4.1.10 Describe about the chemical attack on structural timber

4.2.0 Understand bioconversion of wood

- 4.2.1 Understand the bio-conversion of wood by micro organisms, enzymes and destructive distillation of wood
- 4.2.2 Describe the action of enzymes on wood
- 4.2.3 Describe the enzymes producing fungus
- 4.2.4 Describe the fungal decay of wood
- 4.2.5 Describe the degradation process in wood by enzymes due to micro organisms
- 4.2.6 Discuss the enzymatic hydrolysis by brown rot fungus and white rot fungus
- 4.2.7 Describe the degradation of wood by insects

4.3.0 Comprehend decomposition of wood

- 4.3.1 Define destructive distillation of wood
- 4.3.2 Describe the wood distillation process
- 4.3.3 Describe the wood distillation products such as charcoal, Pyrolignous acid, wood tar.
- 4.3.4 Describe the hard wood and soft wood distillation
- 4.3.5 List various types of wood hydrolysis
- 4.3.6 Describe wood hydrolysis

4.4.0 Understand Wood analysis principles

- 4.4.1 Define wood Analysis
- 4.4.2 State the need of wood analysis
- 4.4.3 Describe the method of preparation of sample for analysis
- 4.4.4 Describe the determination of moisture content, extractives, lignin, ash content
- 4.4.5 Illustrate the apparatus used in the above mentioned process

CONTENT DETAILS

MODULE –I

Organic chemistry - Classification of organic compounds, industrial methods of organic compound, IUPAC – nomenclature of simple organic compound, Industrial methods of preparation, properties and user of hydrocarbons, alkyl and aldehydes, organic alcohol, phenol, ethers, carbonyl compounds, acids and amines, preparation, properties and uses of poly- functional compounds, ethylene glycol, glycerol, oxalic and, lactic acid tartaric acid and salicylic acid. Carbohydrates – structure of glucose, fructose, starch and cellulose, important reaction of these compounds Terpenes – Essential oils, classification of terpenoids, Isolation, Isoprene rule and list the examples of terpenes. Effect of substituents on – acidity of acids & Phenol basicity of amines Hybridization of carbon in methane, ethane, ethylene and acetylene.

MODULE –II

Physical chemistry – theory of corrosion and passivity methods to prevent theory of corrosion PH and its application, introduction to colloids lyophobic and lyophilic sols, their preparation mechanical, electrical and optical properties and uses a brief idea about gels and emulsion Nuclear chemistry - natural radio activity nature of radiation's, disintegration theory, saddy's group displacement law, radio active series, unit of radio activity, rate of radio active decay (no derivation). Artificial or induced radio activity. - Artificial transmission, Radioisotopes and their uses.

MODULE III

Chemistry of wood constituents Composition of wood – cell wall components, cellulose, hemi cellulose, lignin and holo cellulose and carbohydrates. Difference in cell wall composition of soft wood and hard wood Wood extractives. Methods of extraction, steam distillation solvent extraction. Important compounds, essential oils (any three) . Properties and uses. Tannins, flavanoids and dyestuff. Effect of extractives on durability of wood, gluing characteristics, surface coating.

MODULE IV

Wood Hydrolysis and Wood Analysis. Action of – water on wood, aqueous solutions of inorganic salts on wood, alkaline degradation of wood. Oxidation and bleaching of wood. Effect of chemical on wood Destructive distillation of wood, bioconversion of wood by microorganisms and enzymes Wood hydrolysis – Acid hydrolysis Bio- Chemical hydrolysis, enzymatic hydrolysis

REFERENCE

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